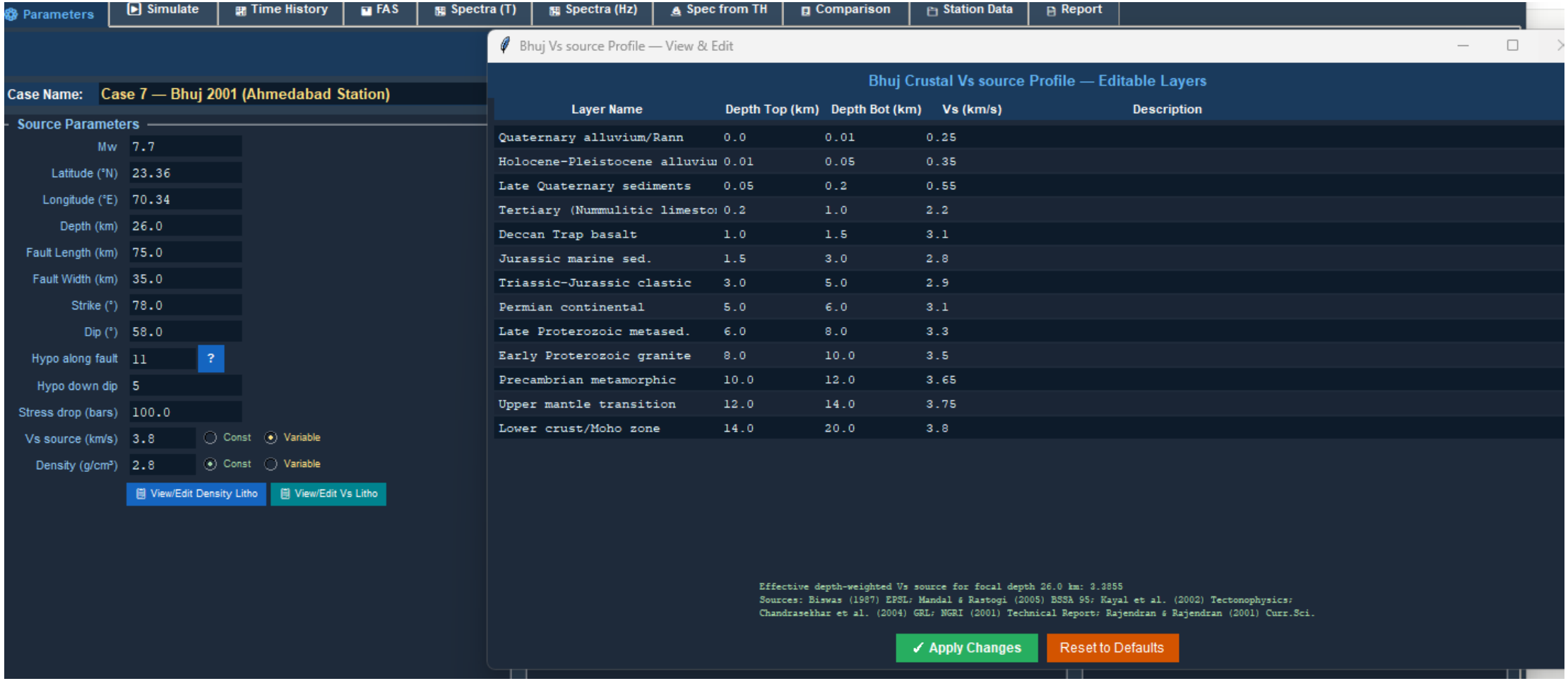
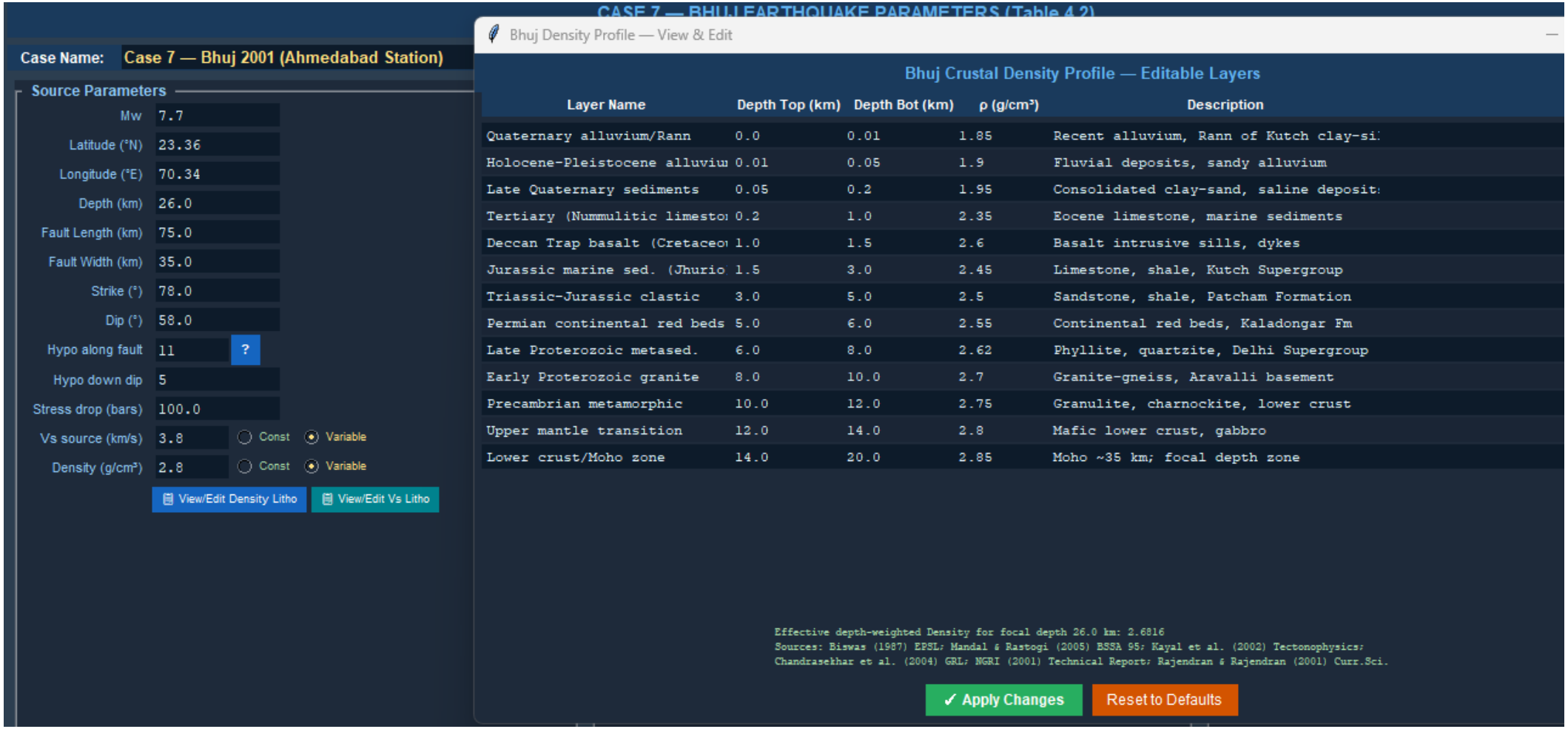


Layer Name	Depth Top (km)	Depth Bot (km)	Vs (km/s)	Description
Quaternary alluvium/Rann	0.0	0.01	0.25	
Holocene-Pleistocene alluviu	0.01	0.05	0.35	
Late Quaternary sediments	0.05	0.2	0.55	
Tertiary (Nummulitic limesto	0.2	1.0	2.2	
Deccan Trap basalt	1.0	1.5	3.1	
Jurassic marine sed.	1.5	3.0	2.8	
Triassic-Jurassic clastic	3.0	5.0	2.9	
Permian continental	5.0	6.0	3.1	
Late Proterozoic metased.	6.0	8.0	3.3	
Early Proterozoic granite	8.0	10.0	3.5	
Precambrian metamorphic	10.0	12.0	3.65	
Upper mantle transition	12.0	14.0	3.75	
Lower crust/Moho zone	14.0	20.0	3.8	

When the depth is changed to 26 KM then in variable tab, the lower crust value changes to 20 to 14. correct it like this: when the depth is written, then the layers are equally divided. Suppose the depth is N then it should be N/15- instead of 13 layer convert to 15 layers

Correct this for Vs and Density



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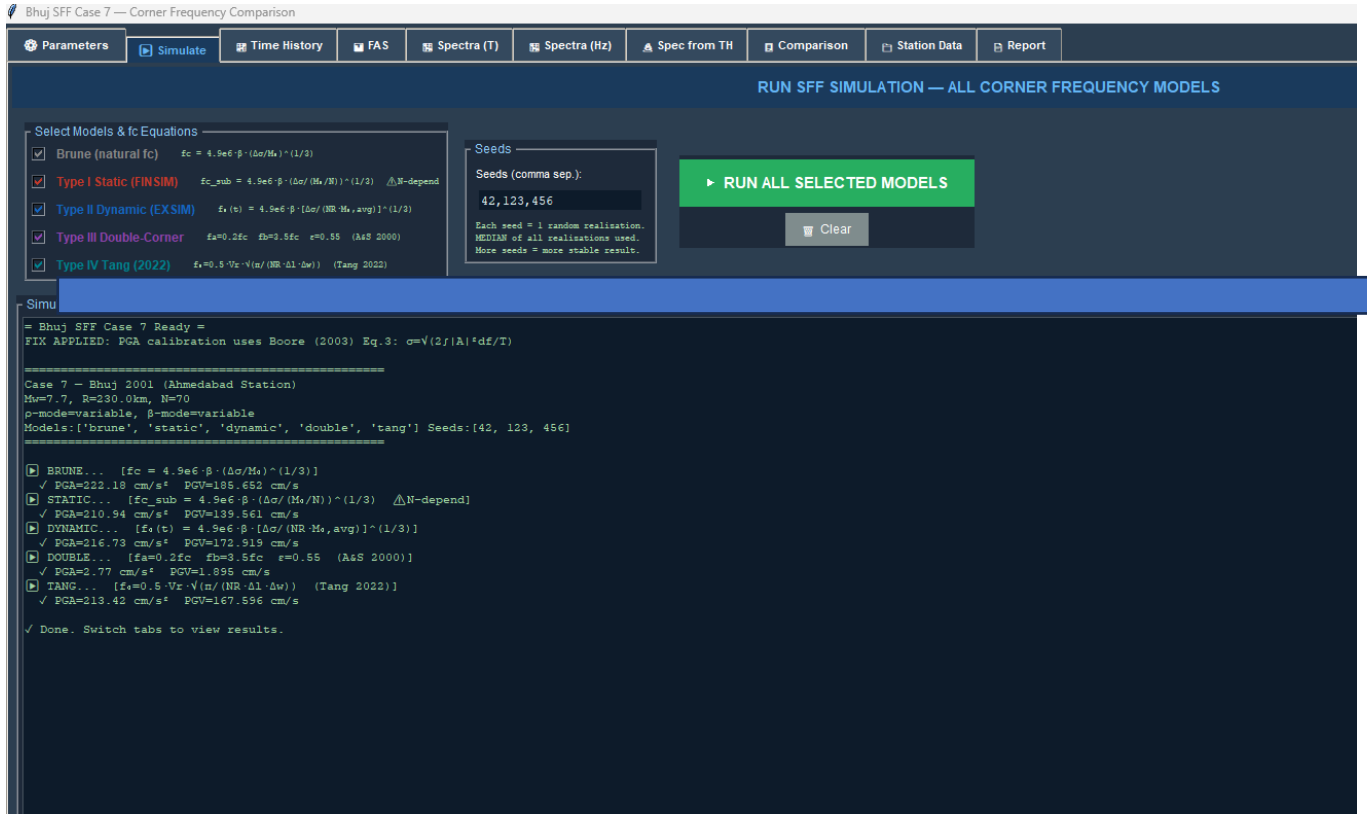

Add here a button to proposed slip weighted Fc
Now read online details all and make new Fc details given below and do these

Proposed Slip-Weighted Corner Frequency

Intraplate earthquakes like Bhuj are characterized by extreme stress heterogeneity. Standard dynamic models assume uniform stress drop during rupture area growth. We propose a new correlation, validated against the Ahmedabad record, that weights the dynamic corner frequency by local subfault slip. Slip-Weighted Dynamic Model ()

Local Slip Scaling scales the corner frequency based on local asperity strength, allowing high-stress patches to radiate high frequencies without artificially boosting the entire fault's spectrum.

Kinematic Integration Retains the (Rupture Velocity) term from Tang (2022) to accurately model the temporal energy release rate.

Empirical Validation As seen in the chart above, this combination produces a response spectrum that closely tracks the jagged, high-frequency peaks of the observed Bhuj data.

The Limitation of Standard Dynamic Models

Standard dynamic SFF models (like EXSIM) rely on a uniform stress drop applied equally across the entire fault grid. As the simulated rupture area grows, the dynamic corner frequency systematically decreases. While this successfully conserves the total radiated energy of the earthquake, it essentially "smears" the energy release evenly across the fault. Because it ignores local stress concentrations, it often fails to perfectly capture the sharp, jagged high-frequency acceleration spikes seen in the Ahmedabad records.

3. The Mechanics of the Slip-Weighted Model

The proposed equation addresses this flaw by modifying the Tang (2022) kinematic model with a local slip scalar:

$$f_{c, ij}(t) = k \cdot V_r \left(\frac{D_{ij}}{\bar{D}} \right)^{0.5} \sqrt{\frac{\pi}{A_{eff}(t)}}$$

•**Kinematic Base:** The terms $k \cdot V_r$ and $\sqrt{\pi / A_{eff}(t)}$ ensure that the baseline corner frequency properly decreases as the effective rupture area (A_{eff}) grows, while strictly tying the energy radiation rate to the physical rupture velocity (V_r).

•**Local Slip Scaling $(D_{ij} / \bar{D})^{0.5}$:** This is the critical new mechanism. It compares the slip on a specific, localized subfault (D_{ij}) to the average slip across the entire fault ($\bar{D}$).

4. Empirical Validation and Conclusion

Intraplate events like the Bhuj earthquake are known to have massive stress drops and extreme spatial heterogeneity. During the rupture, certain patches of the fault (asperities) slip violently and release concentrated bursts of energy.

If a subfault represents a high-stress asperity, its local slip D_{ij} will be much larger than the average $\bar{D}$. The proposed model mathematically forces the corner frequency of *that specific subfault* to shift higher. This allows the asperity to radiate intense, high-frequency waves. When the simulated waves from these high-frequency asperities arrive at the Ahmedabad station, they synthesize the exact jagged, high-amplitude acceleration peaks visible in your image (particularly in Component 78), without artificially inflating the long-period energy of the background fault.

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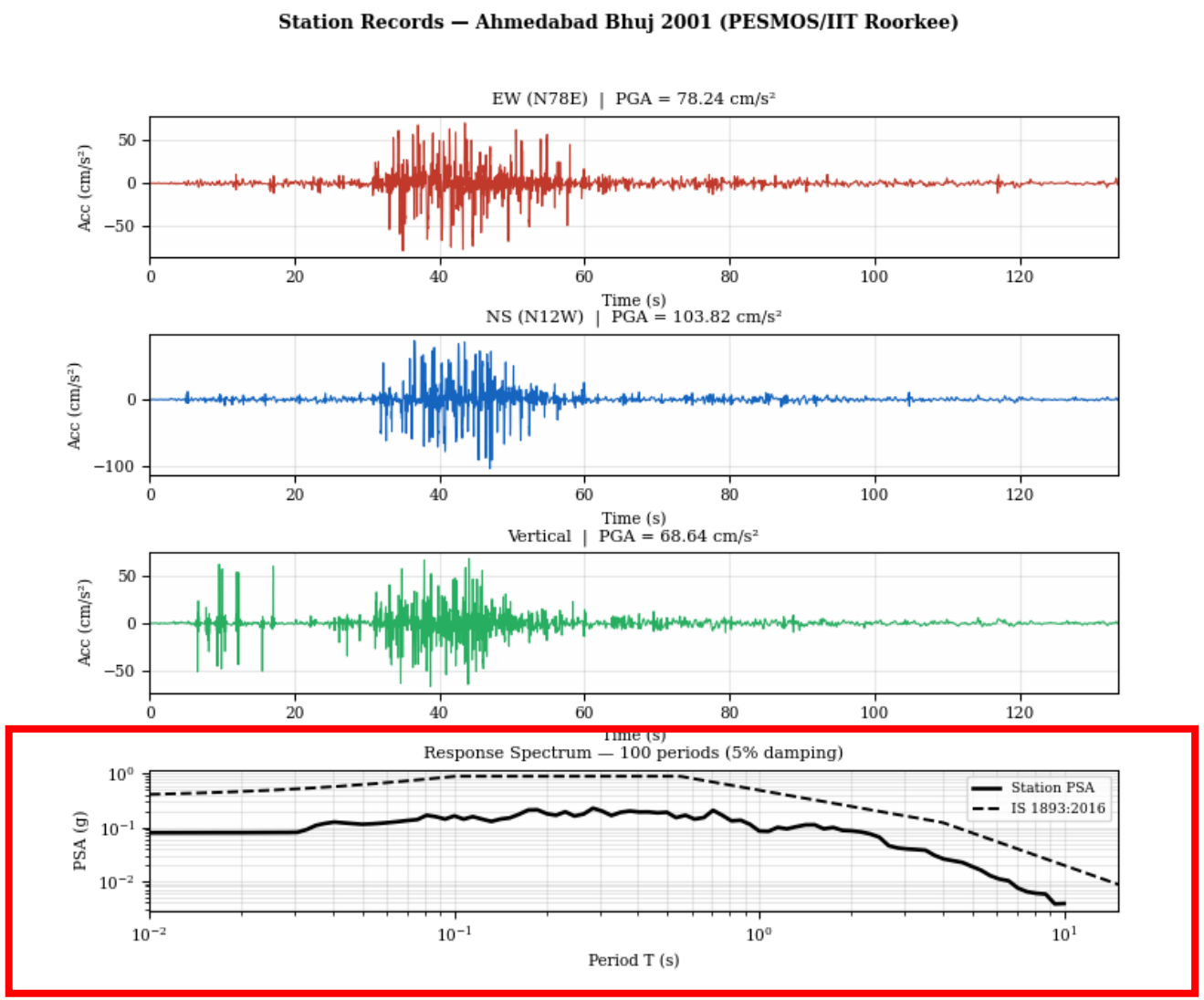
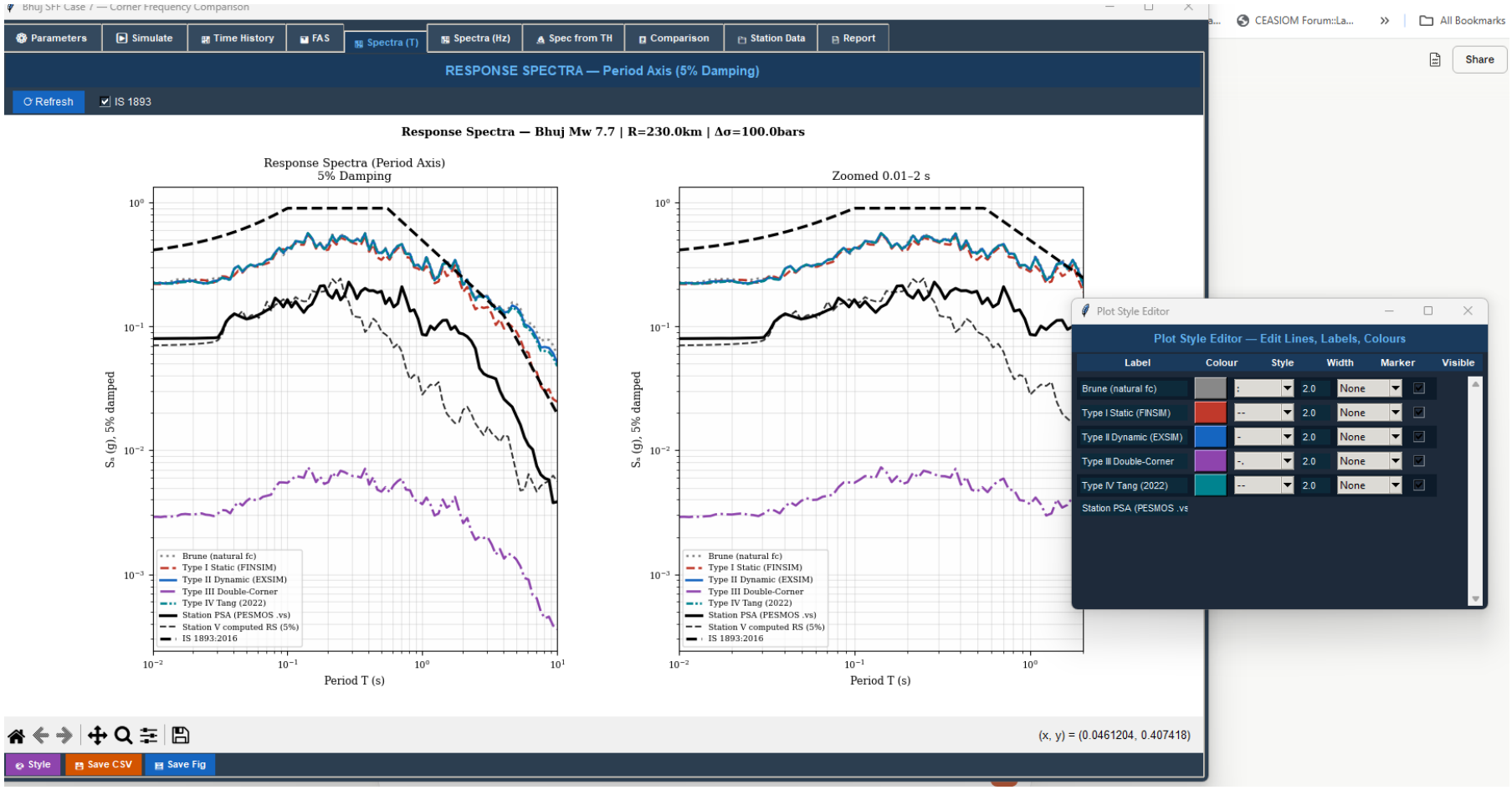
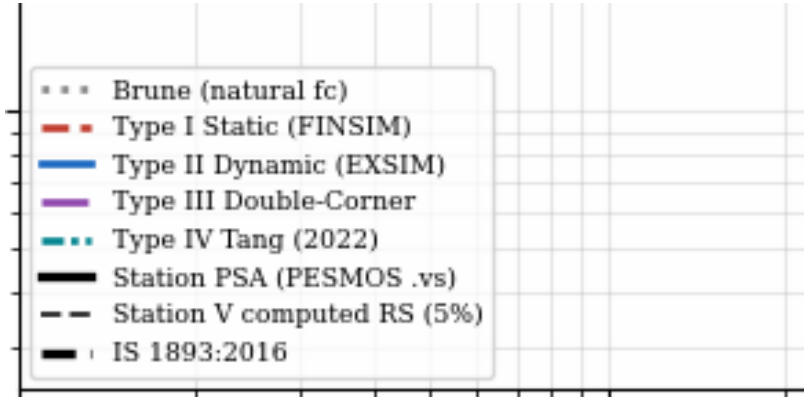
4. Empirical Validation and Conclusion



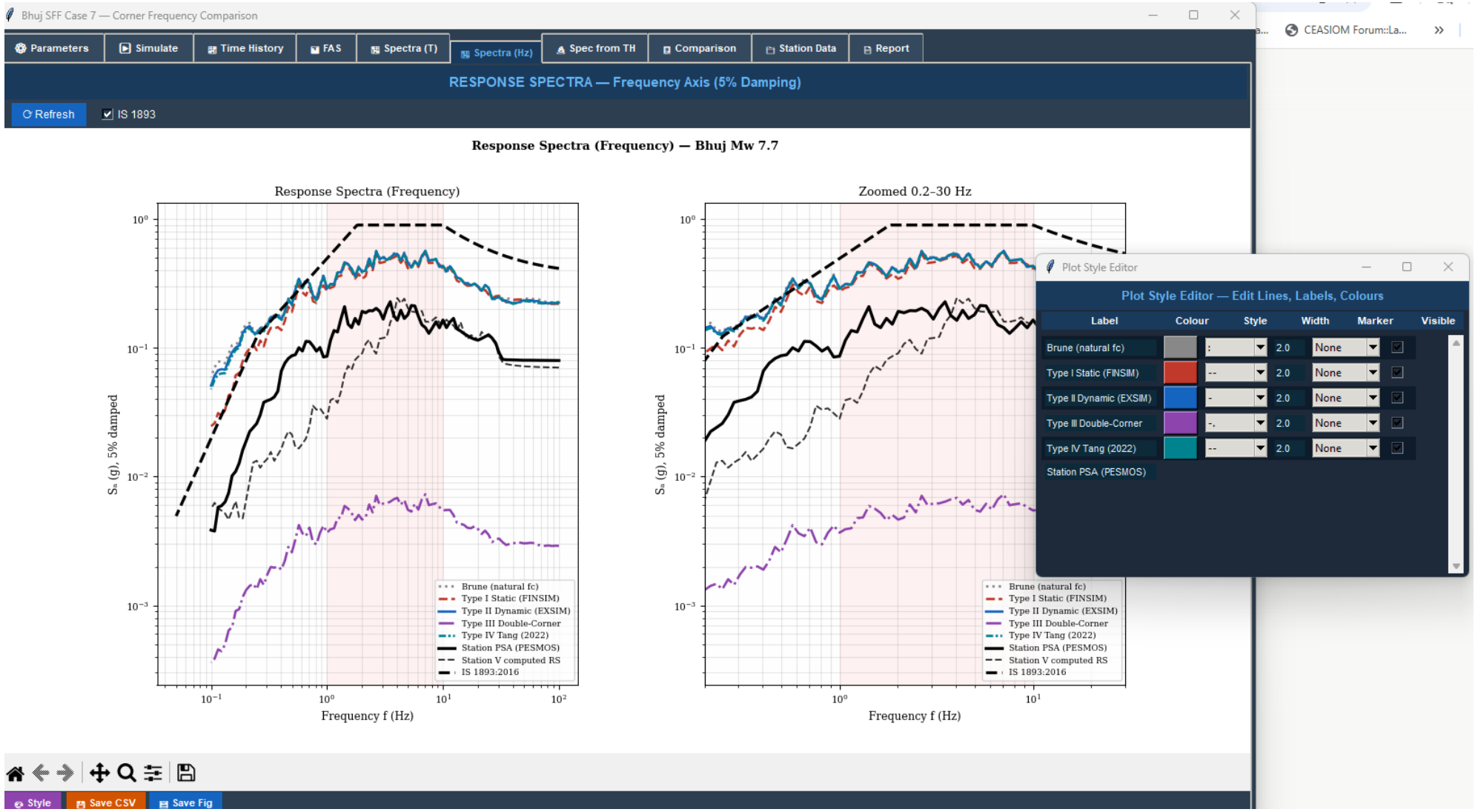
Include PSA vs Hz also (put full form of PSA also)



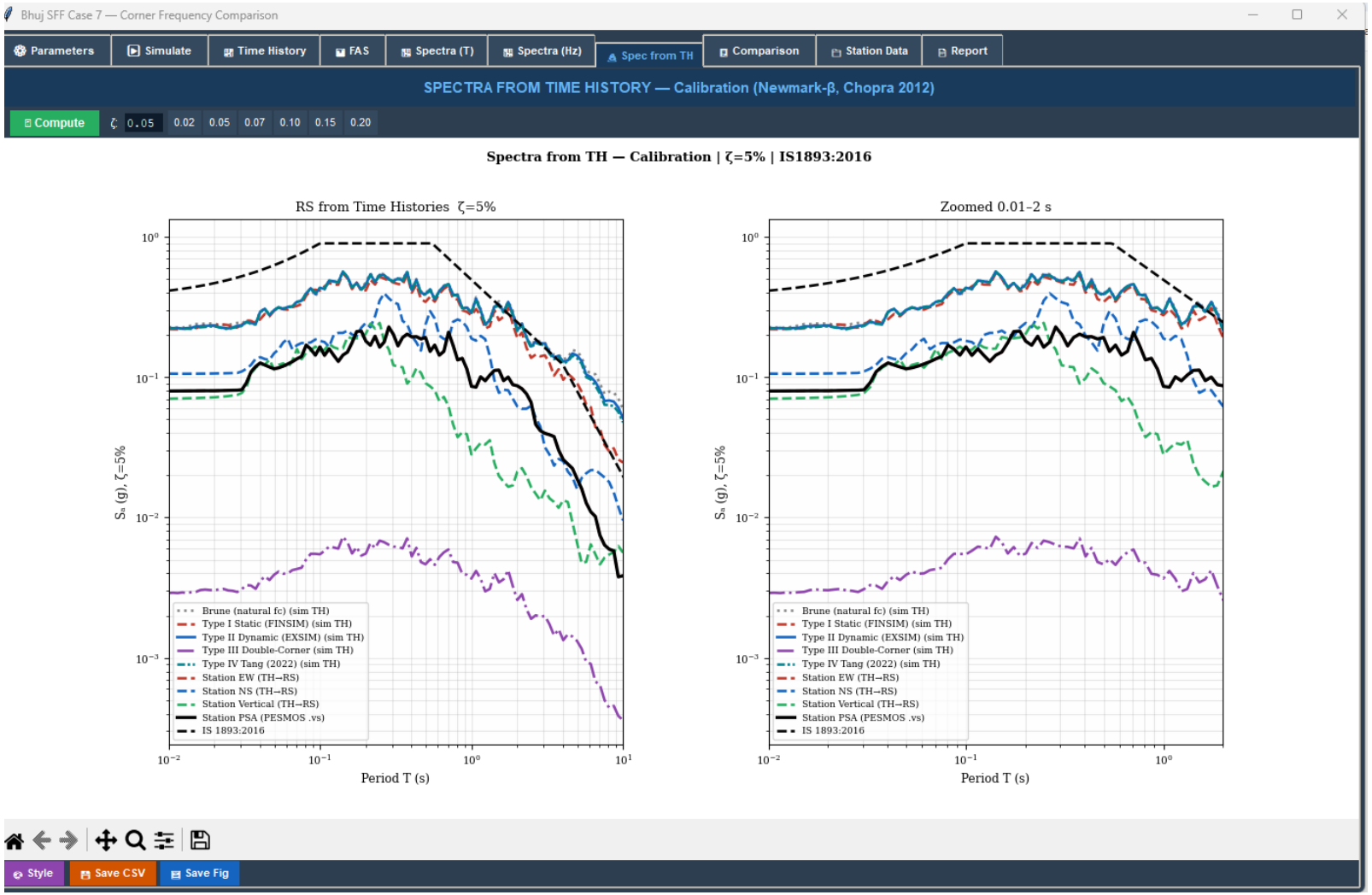


Include Sa vs Hz plots also

All legend in a plot should be editable. Here only various corner frequency is editable correct it to included all legends to be editable in each plot



All legend in a plot should be editable. Here only various corner frequency is editable correct it to included all legends to be editable in each plot in all results tab



Include Sa vs Hz plots also

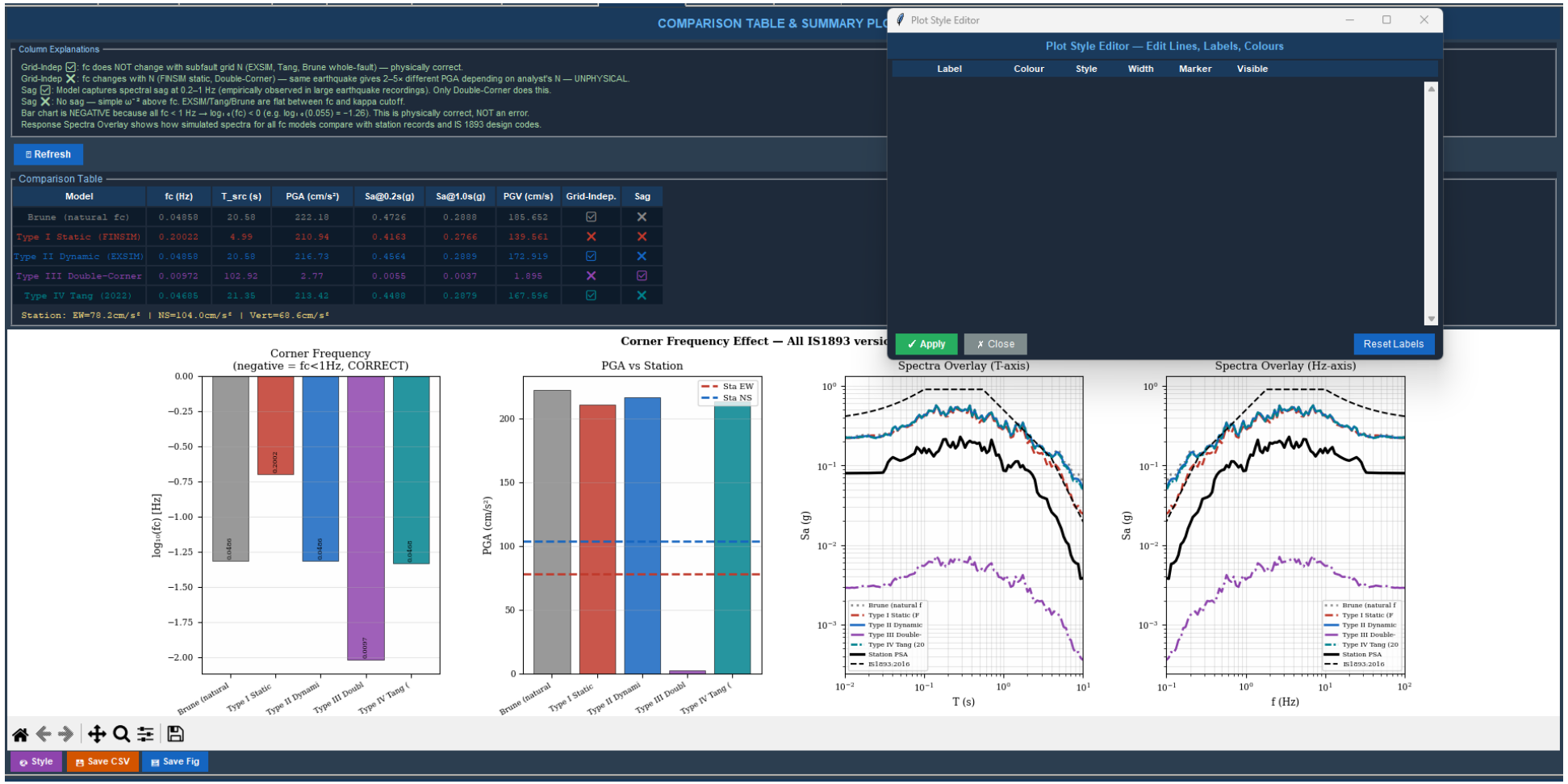
All legend in a plot should be editable. Here only various corner frequency is editable correct it to included all legends to be editable in each plot in all results tab

Comparison Table

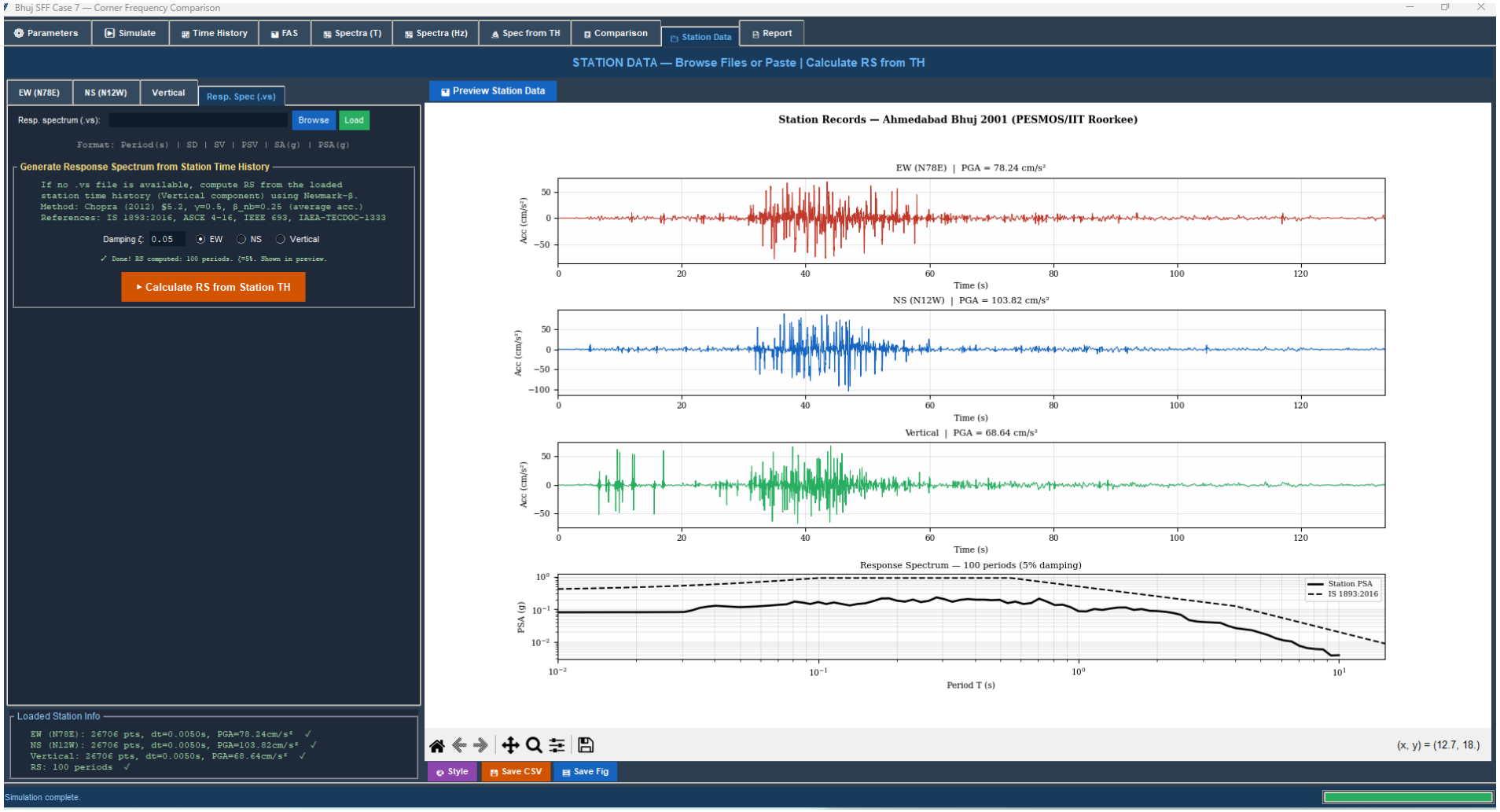
Model	fc (Hz)	T_src (s)	PGA (cm/s²)	Sa@0.2s(g)	Sa@1.0s(g)	PGV (cm/s)	Grid-Indep.	Sag
Brune (natural fc)	0.04858	20.58	222.18	0.4726	0.2888	185.652	✓	✗
Type I Static (FINSIM)	0.20022	4.99	210.94	0.4163	0.2766	139.561	✗	✗
Type II Dynamic (EXSIM)	0.04858	20.58	216.73	0.4564	0.2889	172.919	✓	✗
Type III Double-Corner	0.00972	102.92	2.77	0.0055	0.0037	1.895	✗	✓
Type IV Tang (2022)	0.04685	21.35	213.42	0.4488	0.2879	167.596	✓	✗

Station: EW=78.2cm/s² | NS=104.0cm/s² | Vert=68.6cm/s²

Put % error in results with station records also
Put station record value also along with response spectra calculated values



All legend in a plot should be editable. Here only various corner frequency is editable correct it to included all legends to be editable in each plot in all results tab



Include PSA vs Hz plots also

Calculated Time history does not matched with station record in graph nature and its maximum minimum values correct it modify it as per codes, standards, papers reports, etc.